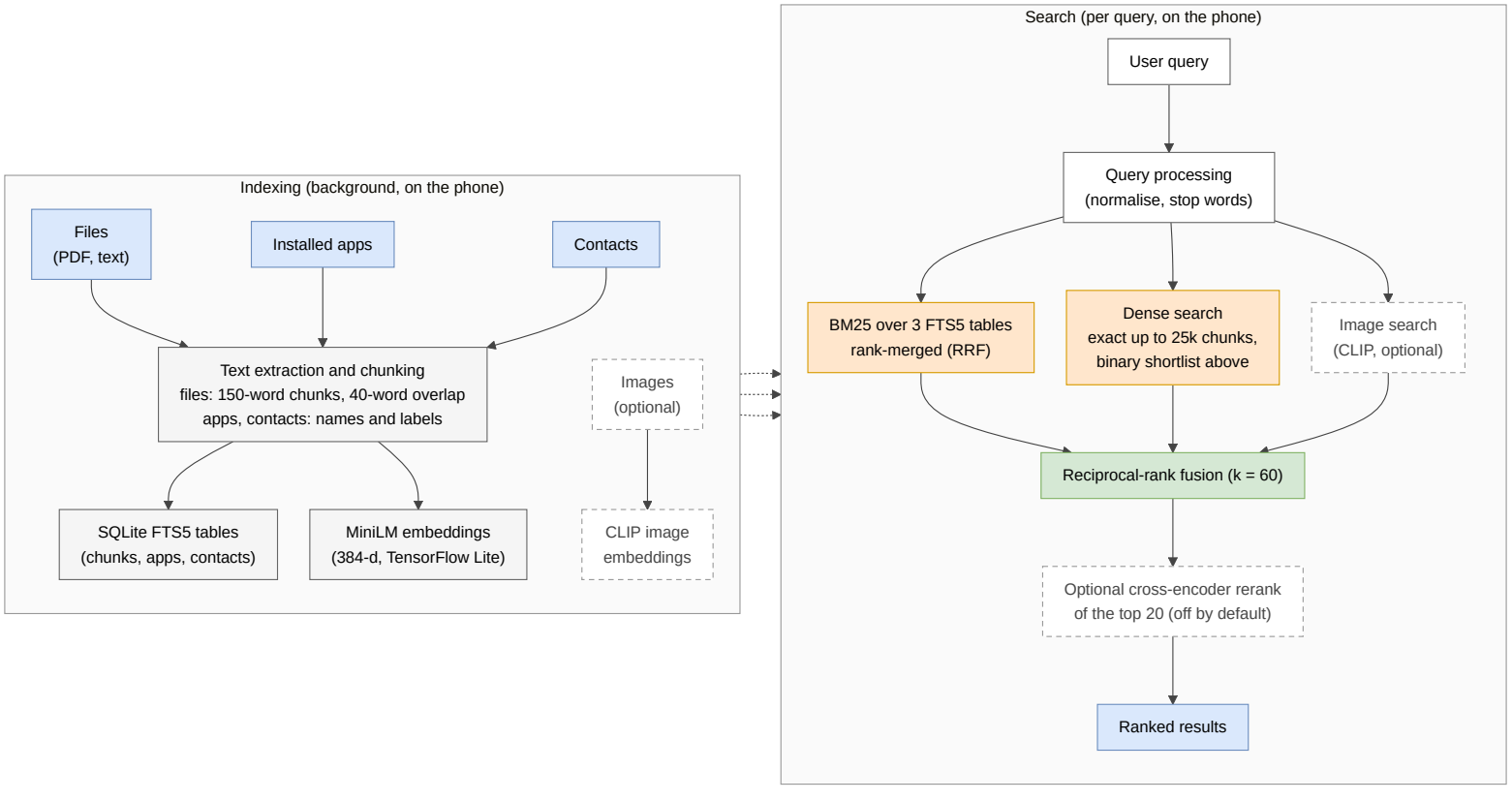
Indexing (background, on the phone)
Files
(PDF, text)
Installed apps
Contacts
Text extraction and chunking
files: 150-word chunks, 40-word overlap
apps, contacts: names and labels
Images
(optional)
SQLite FTS5 tables
(chunks, apps, contacts)
MiniLM embeddings
(384-d, TensorFlow Lite)
CLIP image
embeddings
Search (per query, on the phone)
User query
Query processing
(normalise, stop words)
BM25 over 3 FTS5 tables
rank-merged (RRF)
Dense search
exact up to 25k chunks,
binary shortlist above
Image search
(CLIP, optional)
Reciprocal-rank fusion (k = 60)
Optional cross-encoder rerank
of the top 20 (off by default)
Ranked results